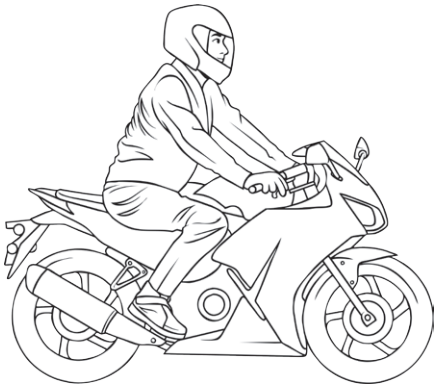
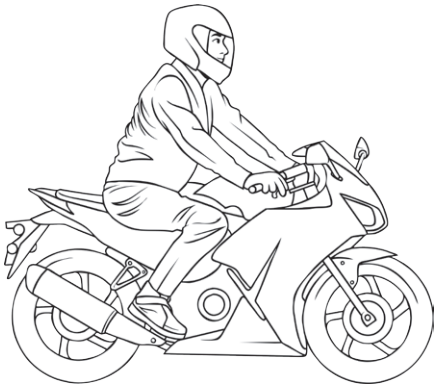
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		Date due in:

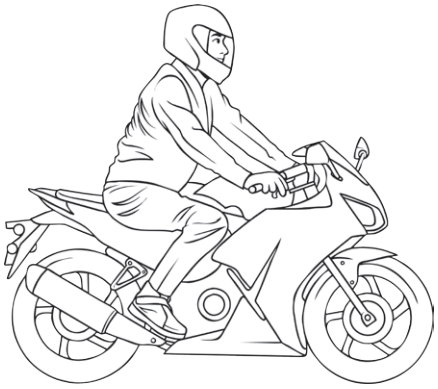
Balanced and Unbalanced Forces



A
accelerating



B
decelerating



c
steady speed

1a. On which motorbike is the force of air resistance greater than the thrust generated from the engine? [1]

1b. On which motorbike is the combined force of air resistance and friction balanced with the engine thrust? [1]

1c. Which force is greatest on motorbike A? [1]

For each description below, state if the forces are balanced or unbalanced. [4]

2a. A train travelling at a steady speed. _____

2b. A ball bouncing off a wall. _____

2c. A horse jumping over a hedge. _____

2d. A submarine descending at a steady speed. _____

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Describe what will happen in each scenario, in terms of forces and movement. Show a force diagram in your answer.

3a. A dodgem (bumper car) is hit by another from the side. [4]

3b. A skydiver releases his parachute whilst free-falling. [4]

Learning Outcomes (tick if achieved)

Q1, 2	I can identify balanced and unbalanced forces.	
Q3	I can describe how forces will affect an object.	

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Balanced and Unbalanced Forces **Answers**

1a. On which motorbike is the force of air resistance greater than the thrust generated from the engine? [1]

B

1b. On which motorbike is the combined force of air resistance and friction balanced with the engine thrust? [1]

C

1c. Which force is greatest on motorbike A? [1]

thrust

For each description below, state if the forces are balanced or unbalanced. [4]

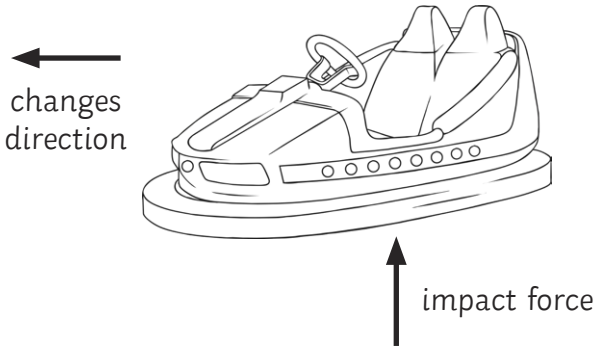
2a. A train travelling at a steady speed.
balanced

2b. A ball bouncing off a wall.
unbalanced

2c. A horse jumping over a hedge.
unbalanced

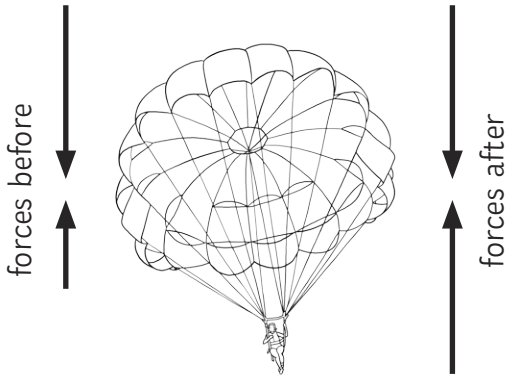
2d. A submarine descending at a steady speed.
unbalanced

3a. A dodgem (bumper car) is hit by another from the side.



The dodgem (bumper car) will change direction because of the force of the impact is greater than the forward force of the dodgem (bumper car).

3b. A skydiver releases his parachute whilst free-falling.



The parachute increases the force of air resistance and the skydiver decelerates. The forces become unbalanced and he changes speed but not direction.